CO1-AT2

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: **ITA 0611** **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I M.Ajay/192312542 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.Ajay

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.
(a) Describe the steps involved in training, validating, and evaluating the model.
(b) Explain how the system can assist in early warning and emergency response.
(5 Marks)

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example. (5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

5. **In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.** (5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

6. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only. (5 Marks)

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

7. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks)

Dataset:

Example	Employment	Salary	Credit Score	Loan History	Loan Approved

M. Ajay
192312542

1. Machine learning system blocks and learning methods

Input Data
↓
Pre-processing
↓
Feature Extraction
↓
Learning Algorithm
↓
Trained model
↓
Prediction / Decision.

Main Blocks:

1) **Input Data**: Increases, text, numbers, Audio etc. are given to the system.

2. **Pre-processing**: Removes errors and prepares the data for learning.

3. **Feature Extraction**: Important features are extracted from the data.

4. **Learning Algorithm**: Learns patterns from the data and creates a model.

5. **Prediction / Decision**: The trained model gives the output for new data.

Unsupervised Learning:

The unsupervised learning, unlabelled data is used the system finds groups or patterns by itself

Unlabelled Data
↓
Learning Algorithm
↓
groups / clusters

Ex: Grouping customers into low, medium and high spending groups using K-means.

2. Machine Learning for Disaster Management

Historical + Real-time Data
↓
Pre-processing
↓
Feature selection
↓
ML Model
↓
Disaster Prediction
↓
Early warning + Response

1. **Model selection:**

* Random Forest / XGB, BOOST, can be used

→ For floods, features like rainfall, river level, soil moisture are used

→ For Earthquakes, seismic waves, ground, vibratio and magnitude are used.

Data preprocessing:

* Remove duplicate and incorrect data

* Handle missing values

* Remove sensor noise

Feature Selection:

Important features are selected using correlation analysis and feature importance to improve prediction accuracy.

Training, Validation and Evaluation:

* Training: Model learns from historical disaster data
* Validation: used to tune and select the best model
* Testing: Model is tested with unseen data
* Evaluate using Accuracy, Precision, recall and F1-score.

(c) Early warning and Emergency Response

* Real-time sensor data is continuously monitored
* If the model predicts high disaster risk and alert is generated.

3. Find-S Algorithm:

Find-S Algorithm considers only positive examples (Yes) and ignores negative examples (No)

Attributes: Outlook, Temperature, Humidity, Wind

Step by Step

Start with the most specific hypothesis

$H_0 = < \phi, \phi, \phi, \phi >$

Positive Example	Values	Hypothesis
Ex: 3	Overcast, Hot, High, Weak.	H_1 = <Overcast, Hot, High, Weak>,
Ex: 4	Rainy, Mild, High, Weak.	Outlook: different
		Temperature: different
Ex: 5	Rainy, Cool, Normal, Weak	Humidity, different

4. **Candidate Elimination Algorithms:**

Let Hypothesis be weather, Temperature, Humi
wind step by step updates

Example 1

S_1 = (Sunny, warm, Normal, weak

G_1 = (?, ?, ?, ?)

Example 2 – positive

compare with S:

* Humidity differs → ?
* wind differs → ?

Therefore

S_2 = (Sunny, warm, ?, ?)

G_2 = (?, ?, ?, ?)

Example 3 – Negative

Specialize G to Exclude this negative example

G_3 = {(Sunny, ?, ?, ?) (?, warm, ?, ?)}

S_3 = (Sunny, warm, ?, ?)

Example 4 – Positive

Temperature differs from S so generalize

S_4 = (Sunny, ?, ?, ?)

(?, warm, ?, ?) in G does not cover

Final version space

Specific Boundary:

S = (Sunny, ?, ?, ?)

General Boundary:

G = (Sunny, ?, ?, ?)

Conclusion:

The Person enjoys the sports when the weather is sunny irrespective of temperature, Humidity and wind.

5. Find-S Algorithm:

Only positive examples are considered negative Examples are ignored

Example	Age	income	student	credit rating	Buys
1	Young	high	NO	Fair	NO
2.	Young	high	NO	Excellent	NO
3.	Young	high	NO	Fair	yes
4.	Young	medium	NO	Fair	yes

Initial hypothesis:

$S = (\emptyset, \emptyset, \emptyset, \emptyset)$

Ex 1: NO → Ignore

$S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Ex 2: NO → Ignore

$S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Ex 3: Yes → take values

S = <Middle, High, NO Fair>

Example 4: Yes → compare with previous

→ Age : Middle ≠ Senior → ?

→ Income: High ≠ Medium → ?

→ Student: NO = NO → NO